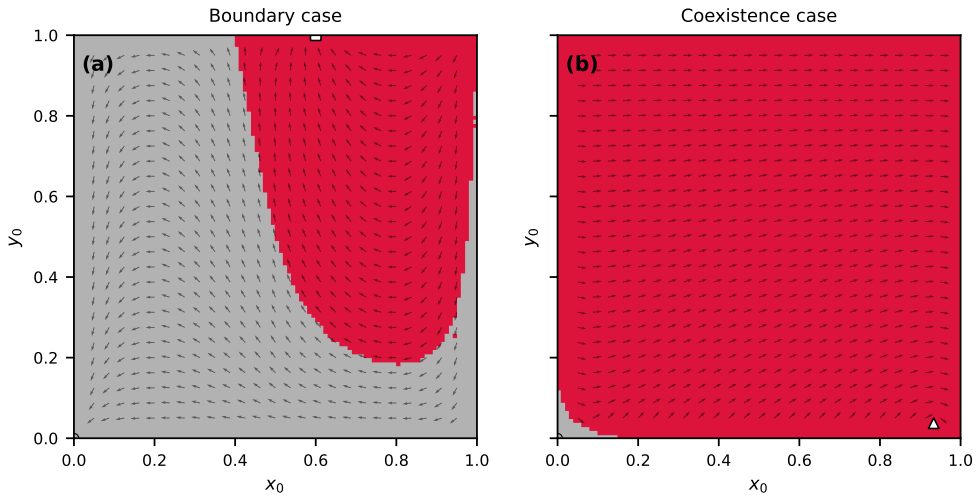

Figure 3: Basins of attraction under two parameter regimes. Left: parameter setting where the boundary equilibrium is stable. Right: setting where the coexistence equilibrium is stable. Red regions indicate initial conditions converging to non-collapsed equilibria, whereas gray regions indicate collapse. Background arrows show the replicator vector field. Basins were computed on a 100×100 grid of initial conditions.

Figure 4 shows this boundary in the $(b_H - b_L, c_A)$ and (σ, c_A) planes. A larger quality gap or lower active-consumption cost reduces the collapse region. The non-collapsed region is maximized near $\sigma = 0.5$ and shrinks as $\sigma \to 0$ or $\sigma \to 1$. In smaller markets with fewer providers (m small), preventing collapse requires stronger support for discrimination (reducing c_A) or a larger quality gap.

Importantly, this collapse boundary is independent of the reward parameters (r, c_H, n) that govern providers' incentives. The first-order condition for sustaining quality lies in the consumers' motivation to differentiate content. Once this condition is met, the specific equilibrium that stabilizes within the non-collapsed domain is determined by the ratio $rn\sigma/c_H$: higher ratios favor the coexistence equilibrium, whereas lower ratios yield boundary-type stability.

These results suggest a two-step design principle for sustaining high-quality digital ecosystems. First, collapse must be avoided by ensuring sufficient discriminability. This can be achieved by (i) reducing active-consumption cost c_A through improved search, labeling, or recommendation systems, (ii) enlarging the quality gap $b_H - b_L$ by suppressing low-quality exposure or penalizing manipulative content, and (iii) optimizing attention capacity σ by limiting excessive information exposure. Second, once collapse is avoided, maintaining coexistence requires increasing the effective reward ratio $rn\sigma/c_H$ by improving incentives for high-effort production or reducing creation costs.

Overall, the model demonstrates that sustainable content quality does not rely on favorable initial conditions but on an institutional balance between discriminability and incentive design.

Discussion

To understand how content quality and audience attention coevolve under limited attention, we developed a minimal theoretical framework that captures the strategic feedback between providers and consumers. The model reveals that content dynamics fall into three regimes separated by a fundamental bifurcation determined by $m\sigma(1-\sigma)(b_H - b_L) = c_A$. This threshold clarifies the role of consumers' discriminability, with $\sigma(1-\sigma)$ indicating that both excessive and insufficient attention undermine quality selection. Thus, simply increasing exposure does not necessarily improve the content environment.

Our contribution is to show that these dynamics emerge from a simple coadaptive mechanism linking selective attention and content effort under explicit attention constraints.

Each regime has a natural interpretation. In the collapse regime, low-quality and non-selective behaviors dominate, resembling environments saturated with repetitive, manipulative, or clickbait content [22, 23, 24]. The coexistence regime represents a dynamic balance in which temporary declines in quality heighten selectivity, restoring providers' incentives and generating self-correcting oscillations. The boundary regime arises when consumers remain selective but

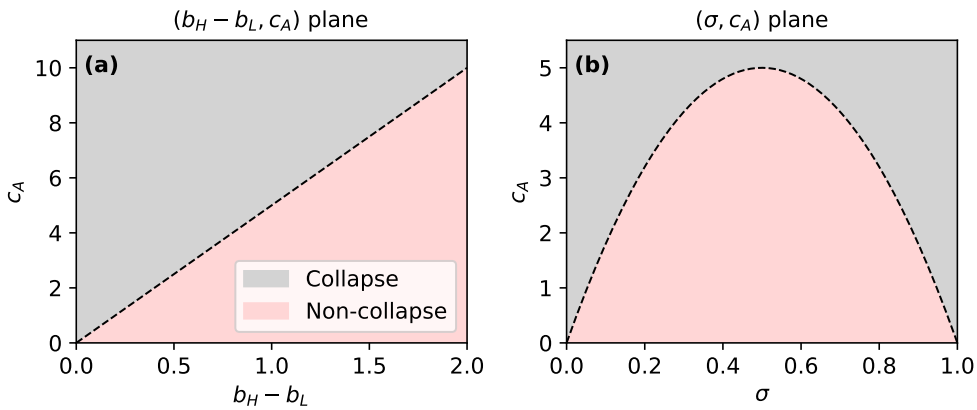

Figure 4: Regimes of collapse and non-collapse. (a) Regimes on the $(b_H - b_L, c_A)$ plane for fixed $\sigma = 0.5$. The dashed line $c_A = m\sigma(b_H - b_L)(1 - \sigma)$ separates collapse (gray, above) from non-collapse (white, below). (b) Regimes on the (σ, c_A) plane for fixed $b_H - b_L = 1$. The non-collapsed region is widest around $\sigma \approx 0.5$ and shrinks as $\sigma \to 0$ or $\sigma \to 1$.

high-effort production is only moderately profitable, a situation characteristic of expert or niche domains. Our basin analysis shows that coexistence is robust to a wide range of initial conditions, whereas boundary outcomes require already favorable levels of selectivity or quality. This asymmetry highlights the importance of interventions that support discriminability before content quality deteriorates.

A related implication concerns recent developments in generative AI. By enabling the rapid creation of large amounts of content, such tools may increase the volume of material competing for users' attention. This can heighten the cognitive effort required to identify reliably high-quality items, effectively raising the functional cost c_A in our model. While this remains a tentative interpretation, it highlights a potential mechanism through which content proliferation may move platforms closer to the collapse boundary when discrimination does not improve accordingly.

These findings offer a two-stage framework for policy and platform design. First, avoiding collapse requires sustaining sufficient discriminability, which can be achieved by lowering cognitive search costs, enlarging the quality gap, or optimizing attention capacity. Importantly, the system performs best in an intermediate range where $\sigma(1 - \sigma)$ is maximized; increasing attention indiscriminately is not always beneficial. Second, once collapse is avoided, stabilizing coexistence requires improving the effective reward ratio $rn\sigma/c_H$ through transparent incentives, lower production costs, or appropriately regulated content throughput. Sustaining content quality thus depends less on initial conditions than on an institutional balance between discriminability and incentives.

The present model abstracts many features of real digital ecosystems. It assumes homogeneous consumers and providers, fixed market size, and simplified functional forms for discrimination and rewards. Relaxing these assumptions could reveal richer dynamics such as multistability or path dependence. Empirically evaluating the model's predictions will require data on selective engagement and content production across platforms. Future extensions may also incorporate cases where low-quality content yields subjective or identity-based utility, such as misinformation, sensationalism, or partisan narratives. Integrating media literacy, trust, and informational health into the model could further clarify how attention and quality interact under algorithmic amplification [36, 37, 38].

Overall, the model indicates that sustaining high-quality content hinges not on favorable initial conditions but on preserving discriminability while maintaining incentives for high-effort creation.

Material and Methods

Replicator formulation

We now develop the analytical formulation of the evolutionary dynamics. Let $x(t)$ denote the fraction of high-effort providers and $y(t)$ the fraction of active consumers. The incentive to invest in effort increases with the prevalence of active consumers, while the value of being active rises with the share of high-quality providers. This bidirectional feedback generates a coevolutionary dynamic process in the pair $(x(t), y(t))$.

In each period, every provider produces one item, and each consumer can attend to a fraction σ of the items produced. We analyze the system at the population level by adopting a mean-field framework, in which providers and consumers are randomly matched and individual fluctuations are smoothed by averaging. The parameter σ thus determines whether the system operates in an attention-saturated or attention-limited regime, depending on whether the fraction of high-quality content x is large enough to meet the attention capacity σ. Accordingly, the payoff structure differs between these regimes, leading to a piecewise-defined form of the expected payoffs that captures the transition between attention-saturated and attention-limited environments. This distinction arises because active consumers may or may not be able to fill their attention budgets with high-quality items, causing the payoff differences to take distinct functional forms across regimes.

Let π denote the expected payoff for each strategy under the mean-field approximation. Subscripts H and L refer to high- and low-effort providers, respectively, while A and P denote active and passive consumers, respectively. Superscripts "prov" and "cons" denote the provider and consumer populations.

(i) Case 1: high-quality supply sufficient ($x \geq \sigma$).

$$\begin{aligned}
&\pi_H^{\text{prov}}(x,y) = -c_H + rn\sigma\left((1-y) + \frac{y}{x}\right), \\
&\pi_L^{\text{prov}}(x,y) = rn\sigma(1-y), \\
&\pi_A^{\text{cons}}(x,y) = -c_A + m\sigma b_H, \\
&\pi_P^{\text{cons}}(x,y) = \sigma m\left(xb_H + (1-x)b_L\right).
\end{aligned} \tag{4}$$

(ii) Case 2: high-quality supply insufficient ($x < \sigma$).

$$\begin{aligned}
&\pi_H^{\text{prov}}(x,y) = -c_H + rn\left(\sigma(1-y) + y\right), \\
&\pi_L^{\text{prov}}(x,y) = rn\left[\sigma(1-y) + y\frac{\sigma - x}{1-x}\right], \\
&\pi_A^{\text{cons}}(x,y) = -c_A + m\left(xb_H + (\sigma - x)b_L\right), \\
&\pi_P^{\text{cons}}(x,y) = \sigma m\left(xb_H + (1-x)b_L\right).
\end{aligned} \tag{5}$$

These expressions are derived by conditioning on whether active consumers can fully satisfy their attention budgets with high-quality items. Detailed derivations are provided in the SI Appendix.

Both providers and consumers adapt their strategies through a finite rationality imitation process. Individuals observe the payoffs of others in their population and switch strategies with a probability proportional to the payoff difference. Thus, the evolution of the population shares is described by a two-population replicator system:

$$\begin{aligned}
\dot{x} &= x(1-x)\left[\pi_H^{\text{prov}}(x,y) - \pi_L^{\text{prov}}(x,y)\right], \\
\dot{y} &= y(1-y)\left[\pi_A^{\text{cons}}(x,y) - \pi_P^{\text{cons}}(x,y)\right].
\end{aligned} \tag{6}$$

As the payoff functions are piecewise defined at $x = \sigma$, the resulting vector field constitutes a piecewise-smooth dynamical system. Substituting the regime-specific payoff functions into the replicator equations yields the following piecewise-smooth system:

$$\dot{x} = \begin{cases} (1-x)\left(rn\sigma y - c_H x\right), & (\sigma \leq x), \\ x\left(rny(1-\sigma) + c_H(x-1)\right), & (\sigma > x), \end{cases} \tag{7}$$

$$\dot{y} = \begin{cases} y(1-y)\left(bm\sigma(1-x) - c_A\right), & (\sigma \leq x), \\ y(1-y)\left(bmx(1-\sigma) - c_A\right), & (\sigma > x), \end{cases} \tag{8}$$

where $b = b_H - b_L$ denotes the benefit differential between the high- and low-quality items. This piecewise structure arises because the payoff differences are represented differently across the two regimes. For $x \geq \sigma$, active consumers are fully saturated with high-quality items, eliminating any dependence on low-quality allocations. For $x < \sigma$, partial substitution toward low-quality items introduces additional linear terms.

Equilibrium points are obtained by solving $\dot{x} = \dot{y} = 0$. In this model, the dynamics yield exactly three stable equilibria: a collapse equilibrium at $(0,0)$, a boundary equilibrium at $y = 1$ with $0 < x^* < 1$, and a single interior coexistence equilibrium. These equilibria represent potential long-run steady states of the evolutionary dynamics. Local stability was evaluated using the eigenvalues of the Jacobian matrix computed at each equilibrium. The explicit equilibrium coordinates and stability conditions are provided in the SI Appendix, and their qualitative implications are discussed in the Results section.